

EXTREMAL GRAPH THEORY

Erdős Problem 915

Maximising Edge Density Subject to Connectivity Constraints

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KU LEUVEN

The title, in plain words

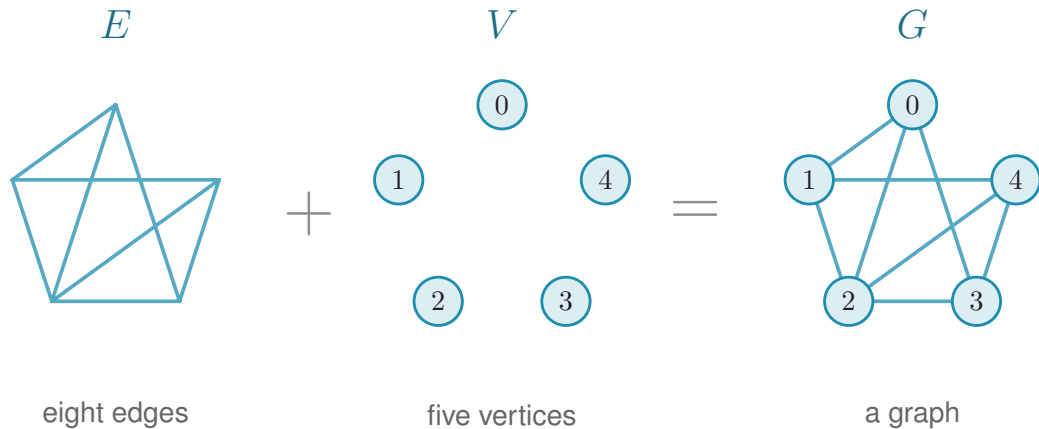
**Maximising edge density in graphs
under connectivity constraints**



translation

networks with many links and low connectivity

A graph is edges and vertices



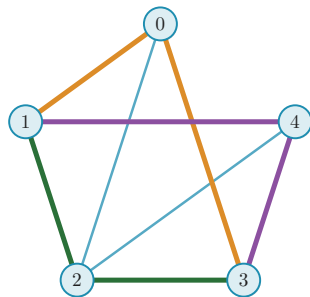
Erdős Problem 915

as many edges as possible, subject to the cap

$$\ell_m(n) = \max_{|V(G)|=n} \{ |E(G)| : \lambda_G^{\max} \leq m-1 \}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v) \text{ over every pair of vertices}$$

$\lambda_G(u, v)$ = the largest number of $u-v$ routes
that share **no edge**: one colour each



$$\lambda(1, 3) = 3$$

three routes from 1 to 3,
three colours, no edge reused

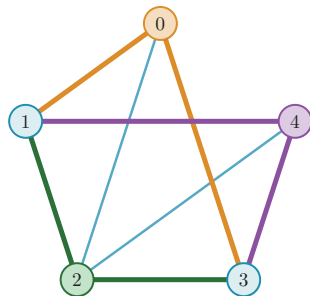
The same question, vertex by vertex

the same cap, read on vertices instead of edges

$$k_m(n) = \max_{|V(G)|=n} \{ |E(G)| : \kappa_G^{\max} \leq m-1 \}$$

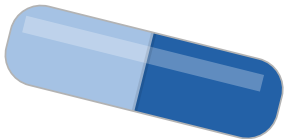
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v) \text{ over every pair of vertices}$$

$\kappa_G(u, v)$ = the largest number of u - v routes that share **no interior vertex**

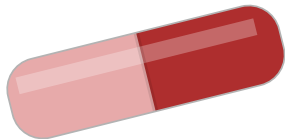
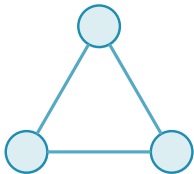


$$\kappa(1, 3) = 3$$

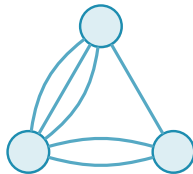
each route keeps its own interior
vertex: 0, 2 and 4

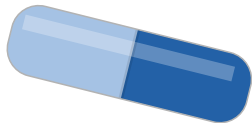


simple

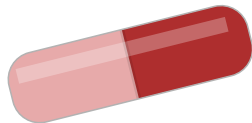
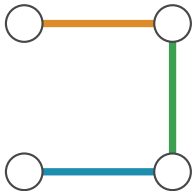


multi

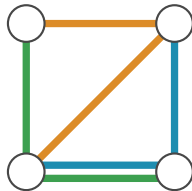


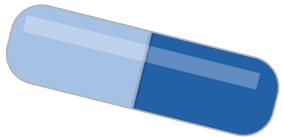


non-hypergraph

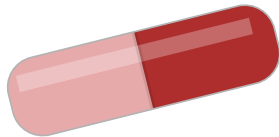
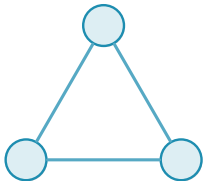


hypergraph

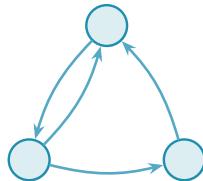




undirected



directed

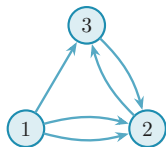


PART II

The program

Sixteen variants, one flow computation.

Modelling: two encodings



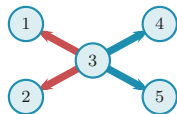
directed multigraph G

$\xrightarrow{\mu}$

$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$

multiplicity matrix μ

$\text{Diag}(\mu) = 0$ (no loops)



directed hypergraph H

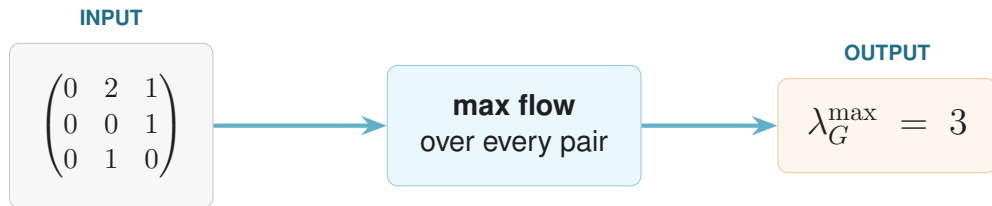
$\longrightarrow$

list of hyperedges:

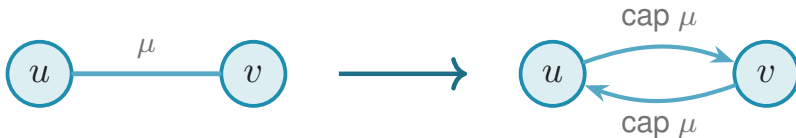
$$e_1 : \{3 \mid 1, 2\}$$

$$e_2 : \{3 \mid 4, 5\}$$

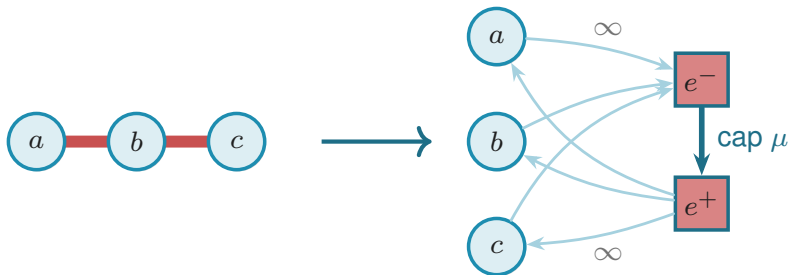
Measuring: max flow over every pair



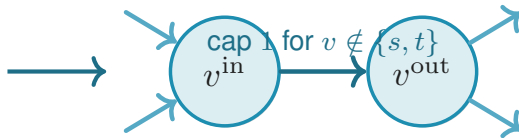
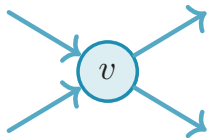
undirected $\rightarrow$ directed



hypergraph $\rightarrow$ non-hypergraph

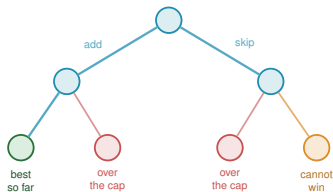


$$\kappa \rightarrow \lambda$$

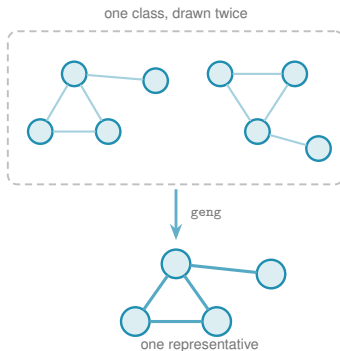


Efficiency: three ways to make the sweep finish

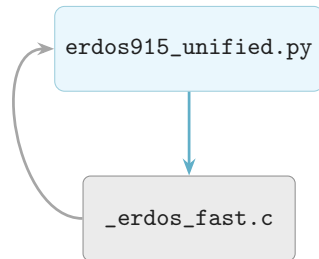
pruning



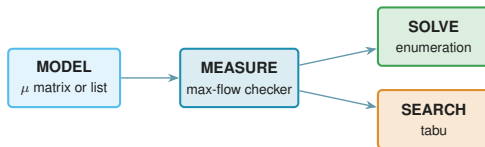
generating search



C: faster max flow



Solving versus searching



solving

- walk every graph on n vertices
- finishes $\Rightarrow$ the exact value
- times out $\Rightarrow$ a lower bound

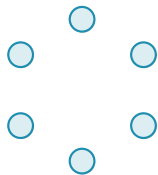
searching

VS

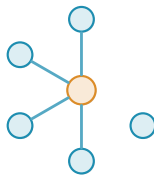
- steer by the energy Φ
- reaches sizes the sweep cannot
- always a lower bound

The search: one number per graph

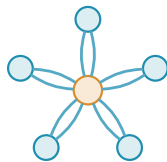
$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \text{discourages excess connectivity}}}$$



$$\Phi = 0$$



$$\Phi = -4$$



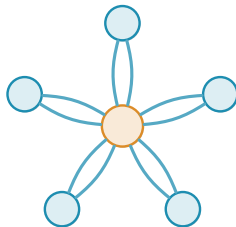
$$\Phi = -10 \text{ at } m = 3$$

Extremal families the search rediscovers



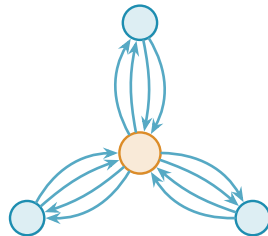
tree

$$n = 6, \ell_2(n) = 5$$



multigraph star

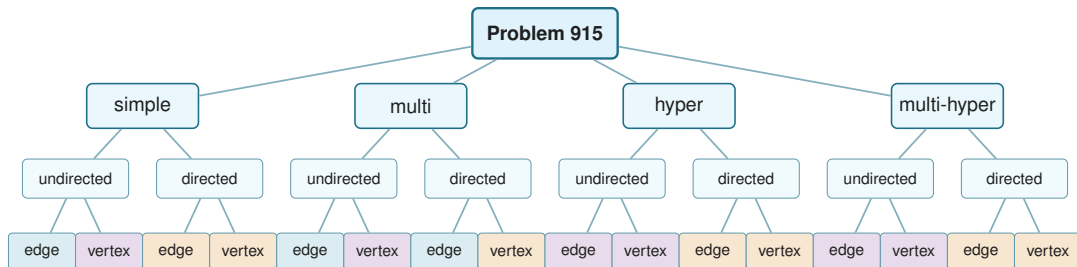
$$n = 6, m = 3, L_3(n) = 10$$



double star

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

Status of the sixteen variants



exact value established in the graph domain



exact value open, asymptotic bounds only



partial results or a conjectural general formula

AI: curse or blessing?

Pros

- $\text{\LaTeX}$ typesetting
- length & complexity of generated text

Cons

- length & complexity of generated text
- control

It's a tool.

short leash + AI proofs = argued conjecture [AI]

Thank you

`github.com/louisivandenbruwaene/thesis`

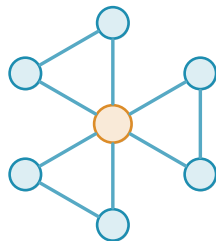
Simple graph · edge

$$\ell_m(n)$$

UPPER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ **PROVED** Mader, via Gomory–Hu

LOWER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ **PROVED** for every $n \geq m$

- the two meet for $n \geq m$, so the value is exact there
- for $n < m$ it is $\binom{n}{2}$: not even K_n carries m routes



hub-and-spokes **EXTREMAL**

COMPUTED

Tabu search returns the tree at $n = 6, m = 2$: $5 = \ell_2(6)$.

Simple graph · vertex

$$k_m(n)$$

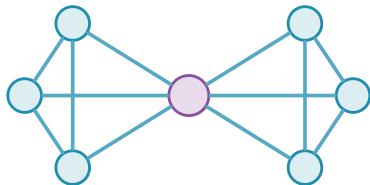
UPPER $k_m(n) = \ell_m(n)$ for $m \leq 4$, $k_5(n) = \lfloor \frac{8n}{3} \rfloor - 4$ for $n \geq 14$ **PROVED**
 $m \geq 6$: **OPEN**

LOWER $c_m > m/2$ for $m \geq 5$ **PROPOSED** under the block formulation

- at $m = 5$ the two agree to $n = 13$, first gap 33 against 32 at $n = 14$
- $c_m = \lim_n k_m(n)/(n - 1)$, and $m/2 < c_m \leq 2(m - 1)$ for $m \geq 5$

COMPUTED

Block sweeps reach small blocks only, and the recursive constructions in the literature show why they cannot fix c_m .



2-connected blocks at a cut vertex

NEAR-EXTREMAL

Simple digraph · arc

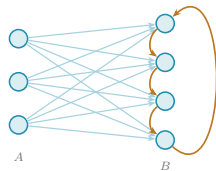
$$\ell_m^{\text{dir}}(n)$$

UPPER $M(n) = \max(2(n-1), \lfloor \frac{n^2}{4} \rfloor)$ at $m = 2$ **PROVED**

$\lfloor \frac{n^2}{4} \rfloor + 4(m-1)(n-1)$ at every n **PROVED**

LOWER $\frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1)$ **PROVED**

- for fixed $m \geq 3$, $\ell_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$; the linear coefficient is open
- the clique-core example disproves the earlier hub-versus-bipartite formula



augmented bipartite digraph

NEAR-EXTREMAL

COMPUTED

Enumeration confirms $m = 2$ through $n = 7$. Tabu returns 6, 10 and 12 at $n = 4, 6$ and 7 .

Simple digraph · vertex

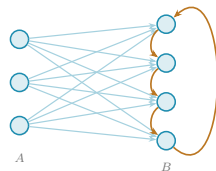
$$k_m^{\text{dir}}(n)$$

UPPER $M(n)$ at $m = 2$ **PROVED**

$$\frac{n^2}{4} + \Theta_m(n) \text{ for fixed } m \geq 3 \quad \textbf{PROVED}$$

LOWER the same bipartite family **PROVED**

- equality with the arc value at $m \geq 3$ is open: an arc upper bound does not constrain the larger vertex-feasible family



augmented bipartite digraph

NEAR-EXTREMAL

COMPUTED

Enumeration confirms $m = 2$ through $n = 7$, and the two separations agree in the completed $m = 3$ runs through $n = 6$.

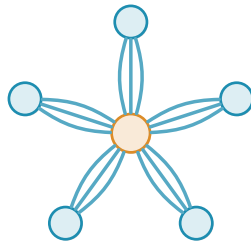
Multigraph · edge

$$L_m(n)$$

UPPER $(m - 1)(n - 1)$ **PROVED** a cut count, any shape

LOWER $(m - 1)(n - 1)$ **PROVED**

- exact for every n and m
- multiplicity removes the simple graph's distance-one restriction



tree, every edge repeated $m - 1$ times

EXTREMAL

COMPUTED

Tabu search returns the multiplicity-2 star at $n = 6$, $m = 3$: $10 = L_3(6)$.

Multigraph · vertex

$$K_m(n)$$

UPPER $K_2(n) = n - 1$, $K_3(n) = 2(n - 1)$ **PROVED**

$$K_m(n) \leq (2 + o(1)) m^2 n$$
 PROPOSED

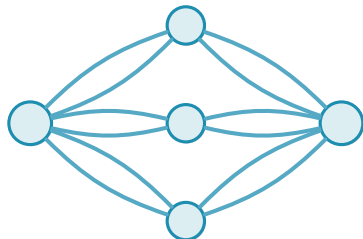
LOWER $K_3(n) \geq 2(n - 1)$ **PROVED**

$$K_m(n) \geq (3 - 2\sqrt{2} + o(1)) m^2 n$$
 PROPOSED

- the two coefficients bracket the growth as $m \rightarrow \infty$ with $n/m \rightarrow \infty$, and no common limit is established
- parallel copies are separated routes, so this is not the simple problem: $K_5(4) = 14$ against $L_5(4) = 12$

COMPUTED

$K_5(4) = 14$ and $K_5(5) = 19$ by exhaustion. A nine-vertex sweep over all 191826 two-connected graphs with at least fourteen edges gives $g_6(9) = 54$.



thickened complete bipartite block

NEAR-EXTREMAL

Directed multigraph · arc

$$L_m^{\text{dir}}(n)$$

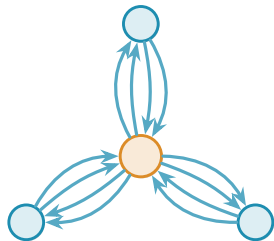
UPPER $(m-1)M(n)$ **PROVED** a sparse skeleton per unit of $m-1$

LOWER $(m-1)M(n)$ **PROVED** two branches, both attained

- exact for every n and m , and the fractional optimum is $M(n)$ too, so one m -free optimisation bounds every threshold
- the classification of the extremisers for $n \geq \max(8, 2m)$ stays **PROPOSED**

COMPUTED

Tabu returns the double star at $m=3$: 12 at $n=4$ and 16 at $n=5$.



thickened bidirected tree **EXTREMAL**

Directed multigraph · vertex

$$K_m^{\text{dir}}(n)$$

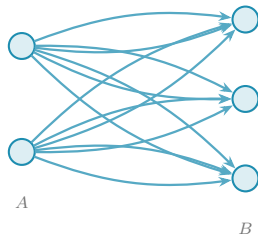
UPPER $(m-1)\frac{n^2}{4} + O_m(n)$ **PROVED**

LOWER $(m-1)\lfloor n^2/4 \rfloor$, thickened bipartite **PROVED**

- the arc extremiser is not even a finite optimum here:
 $K_6^{\text{dir}}(3) = 24$ against an established arc maximum of 20
- the exact correction and the best block shapes are open

COMPUTED

Direct enumeration gives $K_6^{\text{dir}}(3) = 24$.



thickened one-way bipartite digraph

NEAR-EXTREMAL

Simple hypergraph · edge

$$\ell_m^{(r)}(n)$$

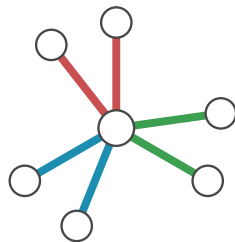
UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ **PROVED** induction on a hyperedge
cut

LOWER the same value **PROVED** when $m-1 \leq \binom{n-2}{r-2}$

- one hyperedge spans r vertices but serves only one route
- outside that range attainment is not guaranteed: at $n = r = m = 3$ the bound reads 2 and only one hyperedge exists

COMPUTED

Exhaustive enumeration gives 11 at $(n, m, r) = (6, 6, 3)$, below the bound 12.



star hypertree ($m = 2$) **EXTREMAL**

Simple hypergraph · vertex

$$k_m^{(r)}(n)$$

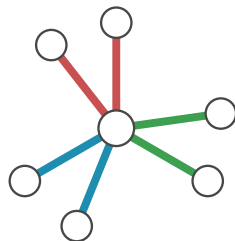
UPPER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ at $m = 2$, $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ at $m = 3$ **PROVED**
 $m \geq 4$: **OPEN**

LOWER a star at $m = 2$, a degree-capped hub at $m = 3$ **PROVED**
 $m = 3$ for $2 < r < n$

- $m = 2$ is feasible exactly when the incidence graph is a forest
- $m = 3$ comes from the incidence-rank inequality
- $\kappa \leq \lambda$ carries an edge bound the wrong way, since vertex feasibility is the weaker restriction

COMPUTED

No counterexample at $m = 4$ across eleven (n, r) cells. The $r = 2$ case of the same problem fails at $m = 5$.



star hypertree ($m = 2$) **EXTREMAL**

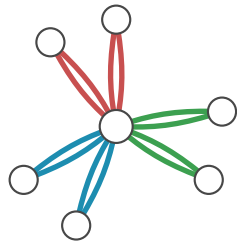
Multihypergraph · edge

$$L_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ **PROVED** unchanged by repetition

LOWER $\frac{(m-1)(n-1)}{r-1}$ **PROVED** when $(r-1) \mid (n-1)$

- the constructions change even though the bound does not
- at $n = r = m = 3$, two copies of the only triple reach a value the simple model cannot



star hypertree, edges repeated $m - 1$ times

EXTREMAL

COMPUTED

A twelve-copy construction at $(n, m, r) = (6, 6, 3)$ exceeds the simple edge optimum of 11.

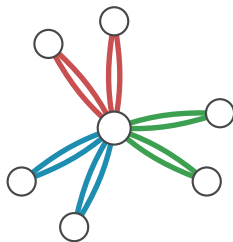
Multihypergraph · vertex

$$K_m^{(r)}(n)$$

UPPER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ at $m = 2$, $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ at $m = 3$ **PROVED**
larger m : **OPEN**

LOWER the repeated star adds attainment cases at $m = 3$
PROVED

- repetition does not help at $m = 2$: two copies of one hyperedge are already two separated routes



repeated star hypertree **EXTREMAL**

COMPUTED

The twelve-copy edge witness at $(6, 6, 3)$ is a vertex lower bound, and does not prove the vertex maximum is twelve.

Directed simple hypergraph · arc

$$\ell_m^{\text{dir}(r)}(n)$$

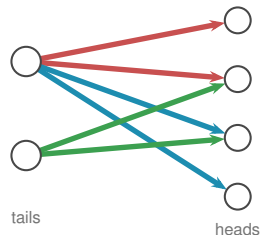
UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED** shadow digraph

LOWER the same constant **PROVED** two-layer, $r \geq 3$

- the two meet, so the leading term is settled and the finite exact values are **OPEN**
- the matching argument covers distinct two-step routes sharing one hyperedge

COMPUTED

At $n = m = 6$, $r = 3$, five distinct outgoing hyperedges per tail give 30, with a direct degree-cut feasibility check.



two-layer construction

ASYMPTOTICALLY EXTREMAL

Directed simple hypergraph · vertex

$$k_m^{\text{dir}(r)}(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED**

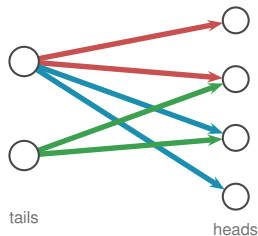
LOWER the same constant **PROVED**

- the shadow argument re-runs under the stricter separation, so the leading term is the same
- whether the finite edge and vertex maxima agree is

OPEN

COMPUTED

Random greedy growth with shuffled restarts measures what the implemented search finds, and establishes no optimum.



two-layer construction

ASYMPTOTICALLY EXTREMAL

Directed multihypergraph · arc

$$L_m^{\text{dir}(r)}(n)$$

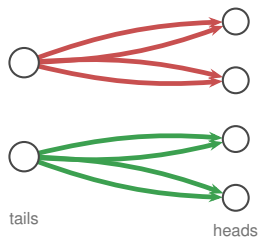
UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED** forward, backward
and general alike

LOWER the same constant **PROVED** repeats lift the
head-set limit

- the finite values can differ from the simple model, and are **OPEN**
- at $r = 2$ these are directed multigraphs, where the simple coefficient is $1/4$ instead, which is why $r \geq 3$ is kept

COMPUTED

The plotted experiments use rank three and the forward convention. General orientations are checked separately in the appendix.



two-layer, hyperedges repeated

ASYMPTOTICALLY EXTREMAL

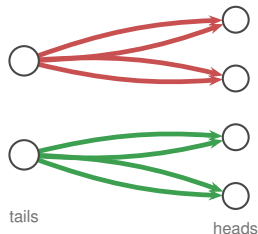
Directed multihypergraph · vertex

$$K_m^{\text{dir}(r)}(n)$$

UPPER $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ **PROVED**

LOWER the same constant **PROVED**

- the same leading term under both separations
- finite exact values **OPEN**, and whether forward and general agree once $n > r$ is open too



two-layer, hyperedges repeated

ASYMPTOTICALLY EXTREMAL

COMPUTED

The rank-three forward-convention runs are not an exhaustive comparison of the three orientation models.